islip science olympiad invitational

Disease Detectives

Do not open this booklet until instructed to do so.
You are allowed to separate the packet and work in any order. The packet must be returned to the event supervisor stapled and in the correct order.

Instructions

You have 50 minutes to complete this exam. Answer all questions completely and thoroughly. You may use one two-sided note sheet and a non-programmable, non-graphing calculator. If you separate this test, write your team name on all sheets.

Team Name_____

Team Number_____

Student Names_____

_____

RAW SCORE

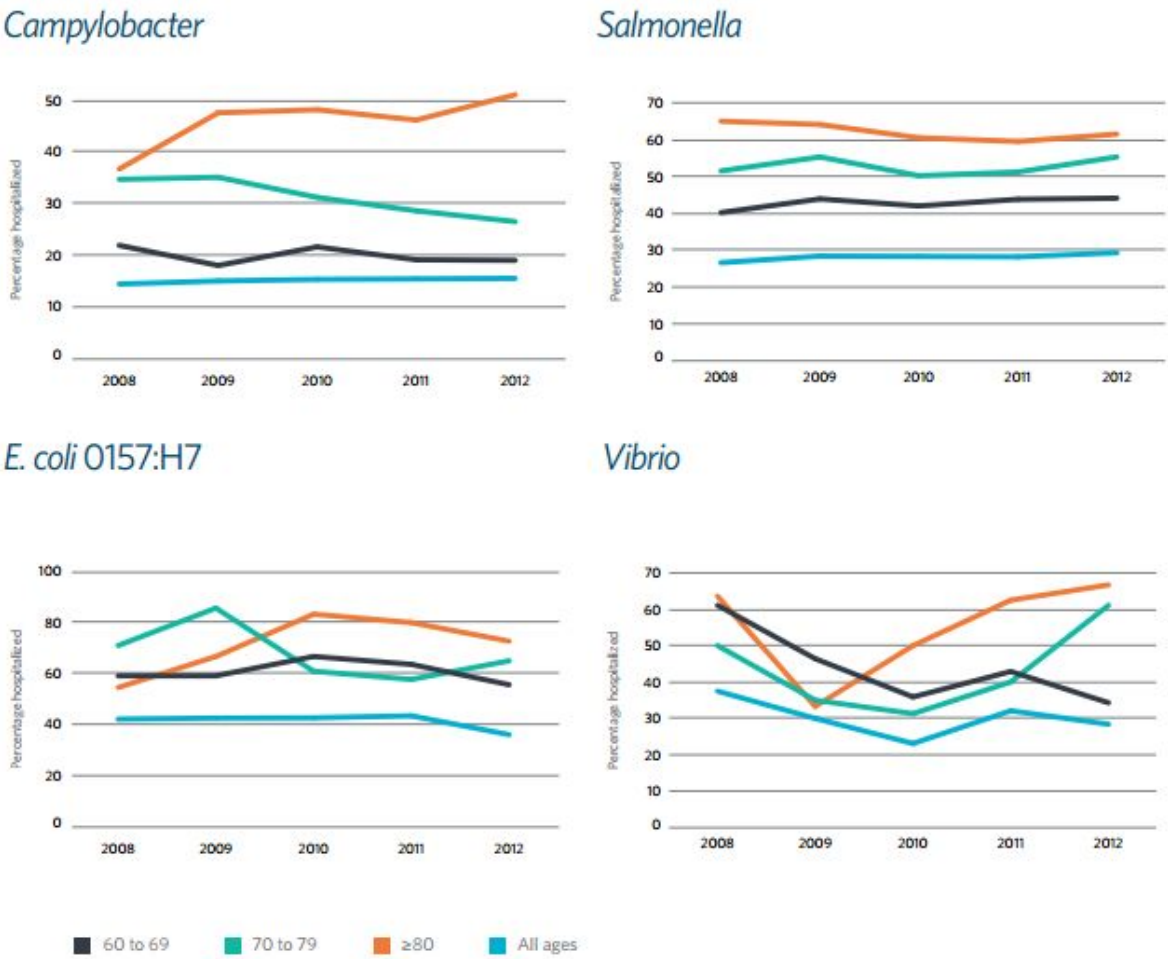
Names: _____ Team:_____

Disease Detectives Exam
Islip Invitational Div. C 2016-2017

1. What is the incubation period for *Bacillus cereus?*

2. Name three modes of disease transmission.

3. Below is the known cases of foodborne illness that exist in different age groups:

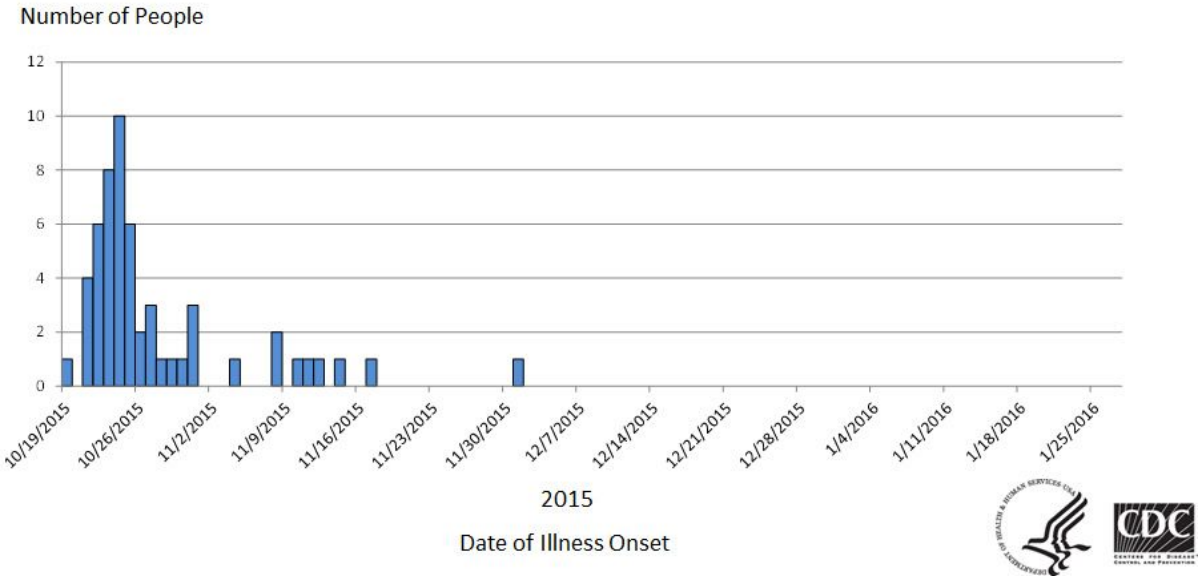


a. For E. coli, what age group has the highest number of people who have been hospitalized in 2011?

b. For Vibrio, what age group has the lowest number of people who has been hospitalized in 2008?

c. Approximately what percent of people between 70 and 79 years of age were hospitalized as a result of Salmonella infection in the year 2010?

4. Below is a graph of the E Coli illness onset in 2015.



The average incubation period for E. Coli is 4 days.

The maximum incubation period for E. coli is 10 days.

The minimum incubation period for E. coli is 2 days.

Based on the Epi Curve and information above,

a. Calculate the average incubation period for this outbreak of E Coli.

b. Calculate the period of exposure for this outbreak of E. Coli.

Names: _____ Team:_____

5.

	Gastroenteritis	No Gastroenteritis
Ate salad	48	12
Did not eat	18	22

a. Calculate the attack rate for the exposed group.

b. Calculate the attack rate for the unexposed group.

c. Calculate relative risk for this data set.

d. Interpret the results from part c.

e. What type of study is this?

f. Name three symptoms of gastroenteritis.

6.

		Case Patients	Controls
Ate Chipotle?	Yes	a = 25	b = 32
Ate Chipotle?	No	c = 14	d = 63

a. Calculate the odds ratio for this data set.

Names: _____ Team:_____

b. Interpret the results from part a.

c.What type of study is this?

d. Is this a retrospective or prospective study?

e. What test would you use if data from multiple age groups had been provided?

Epidemiologists created the following case definition for an outbreak of Hepatitis A:

Discrete onset of symptoms (nausea, and presence of jaundice

This table shows the characteristics of several patients suspected to have acute Hep A following their admittance to a hospital.

Patient#	Date of Report	Date of Onset	nausea	vomiting	anorexia	fever	dark urine	jaundice	IgM HAV
P1	2/5/2010	2/3/2010	0	1	0	1	1	1	+
P2	2/19/2010	2/14/2010	1	0	1	1	1	1	+
P3	2/7/2010	2/2/2010	0	1	0	1	1	1	+
P4	2/5/2010	2/1/2010	1	1	1	0	0	0	+
P5	2/28/2010	2/23/2010	1	1	0	1	1	1	+
P6	2/18/2010	2/14/2010	0	1	1	0	1	0	+

7. What is this type of chart called?

Names: _____ Team:_____

8. Using the original case definition, state how many cases of acute Hep A there are.

9. Based on the chart, create a revised case definition for Hep A.

10. Using this revised case definition, state how many confirmed cases of acute Hep A there are.

11. What does IgM mean?

12. What is IgM HAV?

Questions 13-18 relate to the following case, taken from Korff, Taback and Beard, "A Coordinated Investigation of a Food Poisoning Outbreak," *Ps* 69:9 909-913 Sept. 1952.

On July 11th, 1951, public health officials in Baltimore received news of a suspected outbreak of food poisoning at a local resort. The previous day, a local an insurance company had held a company picnic for 278 employees and their guests. A buffet dinner was served at 6:00 P.M. The food items offered included sliced cooked ham, fried crab cakes, potato salad, sliced tomatoes, heated baked beans, sliced bread, ice cream, cake and hot coffee, served on paper plates. On July 12th, the food service manager at the resort informed the officials that all of the leftover food had been thrown away and the garbage had been removed. One of the resort's two refrigerators was in use but not functioning, with a measured interior temperature of 68 °F. In addition, the work-table surface was cracked, the meat grinder had putrefied meat adhering to it, and there were no convenient hand washing facilities. The water that was used was from the resort's own well, which tested negative for coliform bacteria. The crab cakes were formed by the cook with his hands and were fried in deep fat at approximately 355 °F shortly before they were served. The potato salad had been mixed by the cook with his hands approximately four hours before the salad was served. Information about the picnickers' symptoms and food consumption was

collected via a questionnaire which was distributed to all of the attendees who could be reached. 261 individuals responded, 97 of whom reported becoming ill, primarily with nausea and vomiting.

13. a) Why was the temperature of the fridge taken?

b)Why was the temperature of the fat taken?

For questions 14-16, name 3 risk factors present in the hotel on the day of the picnic and explain why each might contribute to the spread of disease.

14.

15.

16.

17. What does a coliform test measure? What does it indicate?

18. What is the major fecal coliform species that a coliform test may detect is present?

19. Data from the patients surveyed, including the total number of patients that ate each food and the number of patients that ate each food and also fell ill, were collected. What type of study does this represent?

20. Of the employees the responded to the survey, what is the incidence of food poisoning?

Names: _____ Team:_____

Names: _____ Team:_____

Matching:

A. a protein particle that lacks nucleic acid

B. a cell that lack a cellular membrane that encloses DNA

C. bacteria that have a thicker layer of peptidoglycan outside the cell membrane

____

D. how food is handled to prevent foodborne illness

E. pathogen found in untreated water

F. comes in different sizes: spherical, rod and spiral

G. The number of new cases of foodborne illness in a given population during a specified period

H. Insects and ticks, for example

I. A comb contaminated with lice, for example

J. Transfer of harmful substance to food by hands, food contact

K. the ability of a test to correctly identify people who do not actually have a disease

L. the ability of a test to correctly identify people who actually have a disease

M. an occurrence of cases of a disease that affects a large number of people over a wide geographic area

N. an occurrence of cases of a disease that affects a large proportion of the population across several countries or continents

O. the occurrence of cases of disease in excess of what would normally be expected in a defined community, geographical area or time period

P. a number of cases closely grouped over a particular time period regardless of whether there are more cases than expected

Q. describes a condition that is long lasting or recurrent condition

R. describes a disease that originates in a hospital

____

S. ability of a pathogen to infect a susceptible host

T. a measure of the degree of a disease that a pathogen causes

U. ability of an organism to cause disease

V. an agent that causes disease, especially a microorganism

1. Fomite ____

2. Prion ____

3. Cross Contamination

4. Pathogen ____

5. Cluster ____

6. Chronic ____

7. Specificity ____

8. Food Safety ____

9. Norovirus ____

10. Incidence ____

11. Vector ____

12. Epidemic ____

13. Bacteria ____

14. Sensitivity____

15. Infectivity ____

16. Virulence ____

17. Nosocomial ____

18. Gram positive bacteria

19. Prokaryotes ____

20. Pandemic ____

21. Outbreak ____

22. Pathogenicity ____

Names: _____ Team:_____

Tie Breakers:

1. List and define two types of error in data collection other than bias.

_____

_____

_____

2. List and define the two main types of bias, and give an example of each type.

_____

_____

_____

3. List whether the given disease is fungal or bacterial.

Clostridium perfringens: _____

Mucor circinelloides: _____

Shigellosis: _____

Listeriosis: _____

Aspergillosis: _____

Candidiasis: _____

Islip Invitational - Repaired practice edition

Division C | Disease Detectives | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

Name: _____ Date: _____

4. (6 points) Authored replacement for source item Question 4: Solve this case and explain your reasoning: Investigators interview every attendee on a banquet roster after an outbreak and compare illness by food consumed. Identify the design.

Answer: _____
_____
_____

8. (6 points) Authored replacement for source item Question 8: Solve this case and explain your reasoning: Six cases have onset on day 1, eight on day 2, and two on day 3. Their report dates are all day 5.

Answer: _____
_____
_____

repair-case. (4 points) Authored written substitute for unavailable source material: Why not compute risk from an ordinary case-control sample fraction?

Unit: Identifying study designs.
Worked-case data: Investigators interview every attendee on a banquet roster after an outbreak and compare illness by food consumed. Identify the design.

Answer: _____
_____
_____